2 We sometimes have a similar problem with “small” versus “far away.”

3 It is frequently called just Hubble’s Law by those in the astronomy community, but in 2018 the International Astronomical Union voted to officially recognize Lemaître for his contribution by adding him into the name. As a theorist myself, I approve.

4 In the “nearby” universe, where the recession speeds are small, this is just a simple division problem: velocity divided by Hubble Constant equals distance. For more distant sources, it is complicated by the fact that the Hubble Constant isn’t actually constant over all of cosmic time, and the proportionality isn’t a strict proportionality when the speeds are very high. It is safe to assume in general that if anything in cosmology sounds extremely simple, it’s an approximation, a special case, or the ultimate Theory of Everything we’ve all been searching for all our lives. (I wouldn’t bet on Option 3.)

5 Technically, the ball and the Earth are both pulling on each other, because gravity is a two-way street, but the amount of motion the Earth experiences due to the baseball’s gravitational tug is … not much.

6 You might be wondering if we can just measure the expansion now and ten years from now, and see how it’s changed. Unfortunately, our current technology doesn’t allow for measurements this precise, but in the coming decades we might be able to make this comparison.

7 And other minor things like planets and people, but for the purpose of this discussion we can ignore those.

8 To quote the legendary D:Ream, “things can only get better.”

9 Recollapse isn’t impossible, from our current understanding. If dark energy, which we’ll discuss in the next chapter, has especially weird and unexpected properties, it could reverse our expansion. But the evidence so far doesn’t seem to point us in that direction.